

linux Scenario Challenge

Scenario 1 – Company File Organization

You are a **system administrator** setting up directories for a new company project.

Tasks

1. Create a directory called `company`.
2. Inside it create three directories:

```
hr
finance
it
```

3. Go inside the `hr` directory.
4. Create files:

```
employees.txt
salaries.txt
```

5. Write the following lines inside `employees.txt`:

```
Ali
Sara
Ahmed
```

6. Display the contents of the file.

Scenario 2 – Searching Important Data

Your boss wants to know if **Ahmed** is in the `employees` list.

Tasks

1. Search for the word `Ahmed` in `employees.txt`.
2. Count how many lines exist in `employees.txt`.
3. Display only the **first line** of the file.

Scenario 3 – Backup Operation

Before editing files, you must create backups.

Tasks

1. Create a folder called `backup` inside `company`.
2. Copy the file `employees.txt` into the `backup` folder.
3. Verify the copy exists in `backup`.

Scenario 4 – File Permissions

The `salaries.txt` file should only be readable by the owner.

Tasks

1. Add the text
Confidential Salary Data
into `salaries.txt`.
2. Change permissions so:
 - Owner: **read & write**
 - Group: **no access**
 - Others: **no access**
3. Verify the permission change.

Scenario 5 – System Log Analysis

Create a log file and analyze it.

Tasks

Create `server.log` with the following content:

```
login success
login failed
disk error
login success
network error
```

Now:

1. Display the file.
2. Count how many times the word **login** appears.
3. Show only the lines containing **error**.

Scenario 6 – Power User Challenge

Without creating files one by one:

1. Create **10 files at once** named:

```
test1.txt to test10.txt
```

2. Move all `test` files into a folder called `tests`.
3. Display the directory tree structure.